

Daniel Nguyen

Houston, TX | (251) 309-3500 | danielhnguyen307@gmail.com | linkedin.com/in/daniel-nguyen-aa501243

PROFESSIONAL SUMMARY

Senior Full-Stack & AI Engineer with over 10 years of experience architecting high-scale SaaS platforms and autonomous agent systems. Expert in bridging the gap between traditional distributed systems and modern generative AI, building the robust infrastructure required to support real-time Voice AI and financial automation. Proven track record of optimizing low-latency data pipelines, designing resilient event-driven architectures, and shipping production-grade applications that integrate deeply with legacy Facility Management Systems. Specialized in system reliability, developer tooling, and delivering autonomous business logic that drives operational efficiency.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, Bash, HTML/CSS

Frontend: React, Next.js, Vue.js, Angular, Redux Toolkit, React Query, Tailwind CSS, SPA/SSR, Performance Optimization

Backend & APIs: FastAPI, Django, Node.js, Express.js, Spring Boot, REST, OpenAPI/Swagger, GraphQL, gRPC, WebSockets, Server-Sent Events (SSE), Celery, OAuth2/JWT

AI, LLM & Voice: OpenAI API, Function Calling/Tooling, LangChain, RAG, Embeddings, Vector Search, Pinecone, pgvector, Semantic Caching, Prompt/Evals Pipelines, Whisper (STT), Twilio Voice/Media Streams (mulaw)

Data & Messaging: PostgreSQL, Redis, Kafka, MongoDB, SQL Tuning, Indexing/Partitioning, Background Jobs, Idempotency, Distributed Locking

Cloud & DevOps: AWS (ECS, Lambda, RDS, ElastiCache, S3, CloudWatch, IAM, VPC), Docker, Kubernetes, Terraform (IaC), CI/CD (Jenkins, GitHub Actions), Linux, Git

Testing & Reliability: Pytest, JUnit, Mockito, Integration Testing, Regression/E2E Testing, Load/Latency Testing, Feature Flags, Canary Rollouts, Incident Response, Monitoring/Alerting, Distributed Tracing

PROFESSIONAL EXPERIENCE

Swivl | Senior Full-Stack Engineer

Jan 2025 - Jan 2026

- Architected a real-time Voice AI Agent for inbound support, **reducing live-agent routing by 40% QoQ vs. baseline** through autonomous intent resolution.
- Engineered a bi-directional FMS sync engine (StorEdge/SiteLink) with distributed locking, **ensuring idempotent writes across concurrent updates for 50,000+ units**.
- Optimized FastAPI WebSocket ingestion for Twilio Media Streams (base64 mulaw), **enabling seamless barge-in with <500ms end-to-end latency**.
- Implemented Redis Semantic Caching for top intents (e.g., gate codes), **bypassing LLM inference for 35% of calls and cutting p95 latency by 600ms**.

- Developed a secure Action Layer with Stripe Connect for autonomous split-payments, **reducing manual reconciliation churn by directly posting to tenant ledgers.**
- Built a Next.js Live Monitor using SSE for real-time transcripts, utilizing DistilBERT sentiment analysis to flag negative tonal shifts and trigger operator intervention.
- Engineered a telephony emulation suite to mock Twilio streams and FMS state, **enabling end-to-end regression testing of conversation flows without live call costs.**
- Implemented a PII Redaction Pipeline to mask sensitive credit card digits in memory before audio streams reach persistent storage or LLM providers.

Sierra | Senior Software Engineer

Jan 2023 - Dec 2024

- Built the core ReAct event loop in Python with transactional state management, **enabling agents to reliably execute multi-step workflows with automatic retries.**
- Designed multi-tenant PostgreSQL + pgvector schemas, **tuning partial indexes to query millions of rows at single-digit ms p95 latency.**
- Built a React Trace Visualizer to render step-level agent execution (tool calls + retrieval context) for **faster debugging and prompt iteration.**
- Provisioned AWS infrastructure (ECS/RDS) using Terraform, implementing ephemeral environments that **reduced staging spin-up time from days to minutes.**
- Developed a Tool Registry that converts Python type hints into OpenAI function schemas, enforcing strict runtime validation for backend API calls.
- Implemented feature flags and canary rollout strategies to **safely deploy experimental agent behaviors to specific enterprise cohorts without downtime.**
- Created an automated Evals pipeline using Pytest to run regression tests against a golden dataset of conversation scenarios before every deployment.
- Integrated PII redaction middleware to automatically scrub sensitive customer data (SSNs, cards) from prompts before sending to external LLM providers.

HighRadius | Software Engineer II

Jan 2018 - Dec 2022

- Decomposed a monolithic processor into event-driven microservices using Apache Kafka, enabling asynchronous high-volume transaction processing.
- Developed a high-concurrency Java Spring Boot service to parse and normalize unstructured payment data (PDFs, EDI) **at a rate of 10,000+ documents per hour.**
- Engineered a custom React virtualized grid to **render 100,000+ rows with zero performance degradation**, replacing a legacy JSP interface.
- Optimized multi-terabyte SQL queries by implementing table partitioning and covering indexes, **reducing report generation from minutes to sub-seconds.**
- Implemented distributed locking via Redis to ensure data integrity when multiple analysts modify the same invoice simultaneously.
- Collaborated with Data Science to deploy a Propensity to Pay model, building the low-latency inference API for real-time credit risk scoring.

- Wrote comprehensive integration tests using JUnit and Mockito, **increasing code coverage to over 90% and ensuring SOC 2 compliance.**
- Led the migration to a global Redux store, significantly improving data consistency across the financial application suite.

BMC Software | Software Engineer

Jun 2015 - Dec 2017

- Developed a custom Angular directive framework to dynamically render complex ITSM forms based on JSON schemas, enabling no-code workflow building.
- Migrated legacy SOAP endpoints to RESTful Node.js microservices, implementing circuit breakers (Hystrix) to prevent cascading failures.
- Built a log aggregation pipeline using MongoDB, enabling real-time monitoring of server logs and automation success rates.
- Automated build pipelines using Jenkins, **reducing manual release times by 30%** and enabling one-click staging deployments.
- Mentored junior developers on asynchronous JavaScript patterns, conducting code reviews that **improved overall team velocity by 15%.**
- Implemented unit testing frameworks using Jasmine and Karma, establishing the team's first continuous testing protocol for the frontend.

EDUCATION

University of Houston | Houston, TX

Bachelor of Science (B.S.), Computer Science | 2012 - 2015